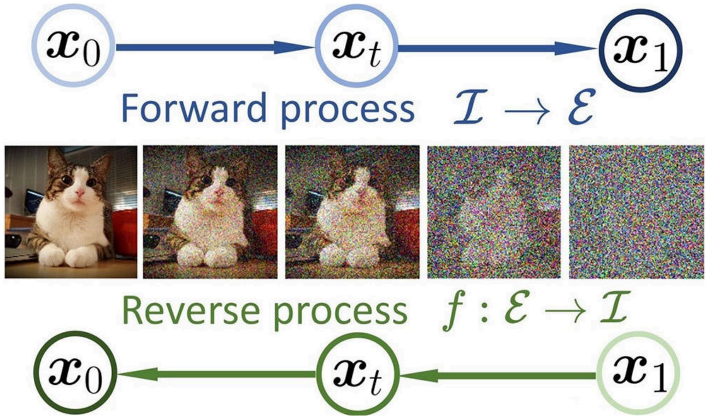
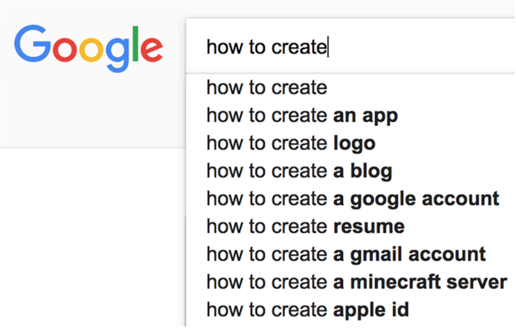
Week 35: Discrete-time Markov chains

Motivation

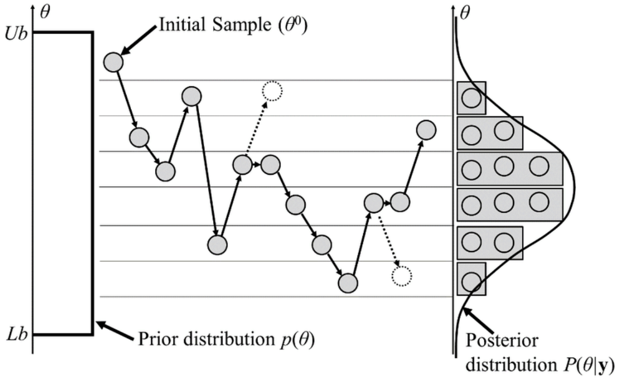


(a) Diffusion models

(b) Drone localisation (HMM)



(c) Language prediction



(d) Monte Carlo and MCMC

Fundamentals

Definition 2.1. *(Discrete-time stochastic process)* A discrete-time stochastic process is a family of random variables $\{X_t : t \in T\}$ where the *index set* T is discrete.

Call X_t the *state* at time t, and the set of all possible states the *state space*.

Definition 2.2. *(Discrete-time Markov chain)* A DT-MC is a discrete-time stochastic process $\{X_t : t \in 0, 1, \ldots\}$ satisfying the *Markov property*

$$P(X_{t+1} = j \mid X_0 = i_0, \ldots, X_t = i_t) = P(X_{t+1} = j \mid X_t = i_t),$$

for $t = 0, 1, \ldots$ and all states $i_0, \ldots, i_t, j$ for which the conditioning event has positive probability.

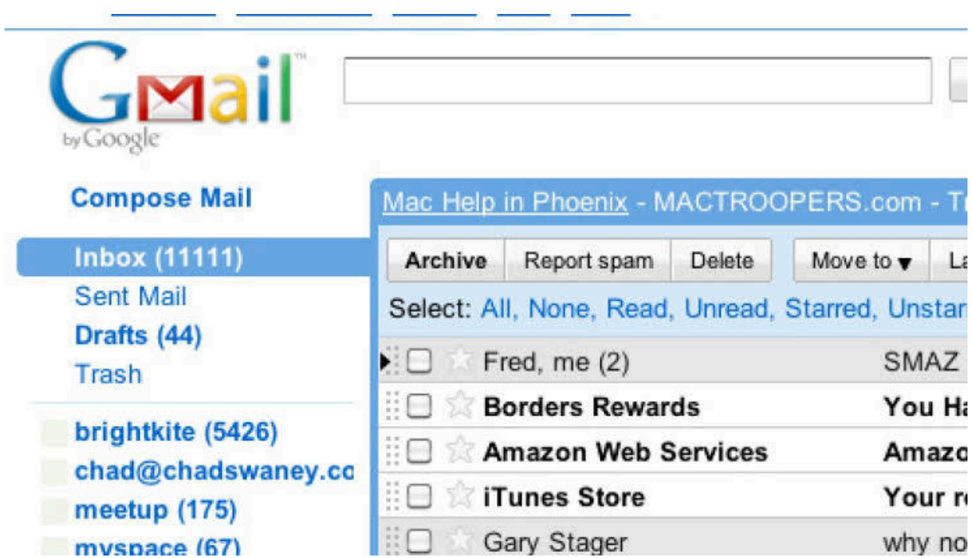
Example of finite state space: Trondheim student

(a) Gløshaugen (G)

(b) Samfundet (S)

(c) Hjem (H)

Example of infinite state space: Receiving and reading emails



Definition 2.3. *(One-step transition probabilities)* For a DT-MC $\{X_t : t \in 0, 1, ...\}$, we call

$$P_{i,j}^{t,t+1} = P(X_{t+1} = j \mid X_t = i)$$

the *one-step transition probabilities* from state i to state j at time t.

Always assume *time-homogeneous* transition probabilities, meaning $P_{i,j}^{t,t+1} = P_{i,j}$ for all $t = 0, 1, ...$ and states i and j.

Definition 2.4. *(Transition diagram)* Let $\{X_t : t = 0, 1, ...\}$ be a DT-MC. A *state transition diagram* visualizes the transition probabilities as a weighted directed graph, where the nodes are the states and the edges are the possible transitions marked with the transition probabilities.

Definition 2.5. *(Transition probability matrix)* For a DT-MC with a finite state space $\{0, 1, ..., N\}$, the *transition probability matrix* is

$$\mathbf{P} = \begin{bmatrix} P_{0,0} & P_{0,1} & \cdots & P_{0,N} \\ P_{1,0} & P_{1,1} & & \vdots \\ \vdots & & \ddots & \\ P_{N,0} & \cdots & & P_{N,N} \end{bmatrix}.$$

Problem 1: From Exercise 2

Consider the DT-MC $\{X_t : t = 0, 1, 2, ...\}$ with state space $\{\mathrm{A}, \mathrm{B}, \mathrm{C}\}$ and transition probability matrix given by

$$\mathbf{P} = \begin{bmatrix} 0.1 & 0.7 & 0.2 \\ 0.5 & 0.1 & 0.4 \\ 0.3 & 0.6 & 0.1 \end{bmatrix}.$$

Knowing that $P(X_0 = \mathrm{A}) = 0.2$, $P(X_0 = \mathrm{B}) = 0.5$ and $P(X_0 = \mathrm{C}) = 0.3$:

- Draw the corresponding transition diagram.
- Compute $P(X_0 = \mathrm{A}, X_1 = \mathrm{B}, X_2 = \mathrm{C})$
- Compute $P(X_1 = \mathrm{A})$
- Compute $P(X_1 = \mathrm{C})$ (here you can use that $P(X_1 = \mathrm{B}) = 0.37$)

Theorem 2.6. *(n-step transition probabilities) For a time-homogeneous DT-MC* $\{X_t : t = 0, 1, \ldots\}$ *with countable state space* S *and any* $m \geq 0$*, we have*

$$P(X_{m+n} = j \mid X_m = i) = P_{i,j}^{(n)} = \sum_{k \in S} P_{i,k} P_{k,j}^{(n-1)}, \quad n > 0, \tag{1}$$

where we define

$$P_{i,j}^{(0)} = \begin{cases} 1 \text{ if } i = j \\ 0 \text{ otherwise} \end{cases}$$

Theorem 2.7. *Recognizing that Equation 1 is a matrix multiplication, we can write*

$$\mathbf{P}^{(n)} = \mathbf{P} \cdot \mathbf{P}^{(n-1)}, \quad n > 0,$$

and iteratively applying this gives

$$\mathbf{P}^{(n)} = \underbrace{\mathbf{P} \cdot \mathbf{P} \cdot \ldots \cdot \mathbf{P}}_{n} = \mathbf{P}^n.$$

Problem 2: From Exercise 2 cont.

Consider the same Markov chain as in Problem 1. We know that

$$\mathbf{P} = \begin{bmatrix} 0.1 & 0.7 & 0.2 \\ 0.5 & 0.1 & 0.4 \\ 0.3 & 0.6 & 0.1 \end{bmatrix}, \quad \mathbf{P}^2 = \begin{bmatrix} 0.42 & 0.26 & 0.32 \\ 0.22 & 0.6 & 0.18 \\ 0.36 & 0.33 & 0.31 \end{bmatrix}, \quad \mathbf{P}^3 = \begin{bmatrix} 0.268 & 0.512 & 0.220 \\ 0.376 & 0.322 & 0.302 \\ 0.294 & 0.471 & 0.235 \end{bmatrix},$$

and that $P(X_0 = \text{A}) = 0.2$, $P(X_0 = \text{B}) = 0.5$ and $P(X_0 = \text{C}) = 0.3$.

Compute the following probabilities:

- $P(X_3 = \text{A})$
- $P(X_6 = \text{A} \mid X_3 = \text{C})$
- $P(X_3 = \text{A} \mid X_1 = \text{B}, X_0 = \text{A})$
- $P(X_3 = \text{C} \mid X_6 = \text{A})$

Problem 3: Exam 2024

A taxi driver provides service in two zones of a city. Fares picked up in zone A will have destinations in zone A with probability 0.6 or in zone B with probability 0.4. Fares picked up in zone B will have destinations in zone A with probability 0.3 or in zone B with probability x.

- Explain that $x = 0.7$.
- Write down the transition matrix and draw the transition probabilities.
- What is the state space? What is the index set of the stochastic process?
- Assume the driver starts in zone A. What is the probability that the driver ends in zone A after two fares? After four fares?

First-step analysis

Definition 3.8. *(Hitting time)* Let $\{X_t : t = 0, 1, ...\}$ be a Markov chain, and let A be a set of states. The *hitting time* of A is the random variable

$$T_A = \inf\{t \geq 0 : X_t \in A\}.$$

We specify $\inf \emptyset = \infty$, meaning $T_A = \infty$ if the chain never reaches A.

Definition 3.9. *(Absorbing state)* For a Markov chain, a state i such that $P_{i,j} = 0$ for all $j \neq i$ is called absorbing. A set of states A is absorbing if $P_{i,j} = 0$ for all $i \in A$ and $j \notin A$.

Problem 4: Analysing absorption

Let $\{X_t : t = 0, 1, ...\}$ be a Markov chain on the state space $\{0, 1, 2\}$. The transition probability matrix is

$$\mathbf{P} = \begin{bmatrix} 1 & 0 & 0 \\ \alpha & \beta & \gamma \\ 0 & 0 & 1 \end{bmatrix},$$

with $\alpha, \beta, \gamma > 0$ and $\alpha + \beta + \gamma = 1$. Assume $X_0 = 1$.

- What is the probability of absorption in state 0?
- What is the expected time until absorption in state 0 or 2? You will need the formula for a geometric series $\sum_{t=0}^{\infty} r^t = 1/(1-r)$

Theorem 3.10. *Let $\{X_t : t = 0, 1, ...\}$ be a Markov chain with state space $S = \{0, 1, ..., N\}$ and transition probability matrix $\mathbf{P}$. Let $A \subseteq S$ be an absorbing set of states.*

- *If u_i is the probability of absorption in $j \in A$ starting from $X_0 = i$, then*

$$u_i = P\left(X_{T_A} = j \mid X_0 = i\right) = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \in A \text{ and } i \neq j \\ \sum_{k \in S} P_{i,k} u_k & \text{otherwise.} \end{cases}$$

- *If $v_i = \mathbb{E}[T_A \mid X_0 = i]$ is the expected time until absorption starting from $X_0 = i$, then*

$$v_i = \begin{cases} 0 & \text{if } i \in A \\ 1 + \sum_{k \in A^{\mathrm{C}}} P_{i,k} v_k & \text{otherwise} \end{cases} \tag{2}$$

Towards long-term behaviour…

$$\mathbf{P} = \begin{bmatrix} 0.8 & 0.2 \\ 0.3 & 0.7 \end{bmatrix}, \quad \mathbf{P}^2 = \begin{bmatrix} 0.7 & 0.3 \\ 0.45 & 0.55 \end{bmatrix}, \quad \mathbf{P}^{10} \approx \begin{bmatrix} 0.600391 & 0.3996090 \\ 0.599414 & 0.400586 \end{bmatrix}, \quad \mathbf{P}^{100} \approx \begin{bmatrix} 0.6 & 0.4 \\ 0.6 & 0.4 \end{bmatrix}.$$